Lifeline Logistics: Mining the Global HIV/AIDS Medicine Supply Chain

Nov 5, 2026 · @Curious

Executive summary

This project looks at about 10,300 real shipments of HIV/AIDS drugs and test kits that the USAID-funded Supply Chain Management System (SCMS) sent to roughly 40 countries. It asks one business question: **which vendors, routes and shipment modes deliver life-saving medicine on time and at the lowest freight cost?**

Most student projects use retail sales, Netflix, Airbnb or churn data. This topic is a global-health procurement chain, an area few people have looked at, and the data is good for data mining because it is messy in useful ways. It has six date fields to mine for delays, freight costs stored as free text ("Freight Included in Commodity Cost", "See ASN-93"), and a natural nested structure (shipment, product, vendor, logistics) that suits a document database.

What the project delivers

1. **MongoDB database** (`scms_lifeline`): cleaned, nested shipment documents plus lookup collections, with indexes and aggregation pipelines for every KPI.
2. **Data mining layer** (Python): delay classification, freight-cost regression, country/vendor clustering and association rules.
3. **Power BI dashboard** (5 pages): Executive overview, Delivery performance, Freight and cost, Vendor and product, Country risk map.
4. **This report**, which documents the design, the methods and the findings as each part is built.

Business problem and objectives

A late or overpriced shipment of antiretrovirals means patients go without treatment, so the procurement team needs to know where delays and freight costs come from. Today that knowledge is spread across six date fields and free-text cost notes, and nobody can see it at a glance.

Stakeholders

Stakeholder	What they decide	What they need from this project
Procurement / sourcing manager	Which vendor and fulfilment route to use	Vendor on-time rate, price per unit, freight share of value
Logistics manager	Air vs truck vs ocean, consolidation	Delay and cost by shipment mode, weight and country
Country programme lead	Safety-stock levels	Delay risk and lead time per country
Donor / finance (USAID)	Budget allocation	Spend by country, product group and year

Research questions

1. **RQ1 Delivery reliability:** What share of shipments arrive after the scheduled date, and which country, vendor, mode and product combinations drive that?
2. **RQ2 Lead time:** How long does it take from purchase order to delivery, and how much of that is vendor time versus transit time?
3. **RQ3 Freight cost:** What drives freight cost per kg, and can it be predicted from weight, mode, country and value?
4. **RQ4 Segmentation:** Which groups of countries and vendors behave alike (low cost and reliable vs high cost and late)?
5. **RQ5 Patterns:** Which combinations of attributes (for example *Air + Direct Drop + Nigeria*) frequently occur together with late delivery?
6. **RQ6 Spend:** How is commodity spend distributed across product groups, molecules and years, and how concentrated is it in a few vendors?

Measurable objectives

- Build a late-delivery classifier with ROC-AUC of at least 0.75 on held-out data.
- Build a freight-cost model that beats a mode-only baseline on MAE.
- Answer every research question on a dashboard page that refreshes from MongoDB.

Dataset

The dataset is **Supply Chain Shipment Pricing Data** (Kaggle), originally published by USAID as the SCMS Delivery History dataset (USAID / AmeriGEOSS catalogue). The file is `SCMS_Delivery_History_Dataset.csv`, 3.8 MB, **10,324 rows x 33 columns**, and each row is one line item of a shipment.

Fact	Value (checked against the file)
Delivery dates covered	2 May 2006 to 14 Sep 2015
Destination countries	43
Vendors	73 (one is "SCMS from RDC", the regional distribution centres)
Distinct products	184 items, 86 molecules / test types, 5 product groups
Projects / ASN-DN shipments	142 project codes / 7,030 shipment notices
Commodity value	US$ 1,627.6 M
Freight cost (numeric rows only)	US$ 107.9 M
Licence	CC BY-ND 4.0 (USAID); we do not redistribute changed data, only our analysis

Data dictionary (raw columns, grouped)

Group	Columns	Type	Notes
Identifiers	ID, Project Code, PQ #, PO / SO #, ASN/DN #	text / int	PQ # is "Pre-PQ Process" in 2,681 rows
Destination	Country, Managed By	category	43 countries; 99.4% managed by PMO - US
Fulfilment	Fulfill Via, Vendor INCO Term, Shipment Mode	category	From RDC 5,404 / Direct Drop 4,920; modes Air, Truck, Air Charter, Ocean; mode missing in 360 rows
Timeline	PQ First Sent to Client Date, PO Sent to Vendor Date, Scheduled Delivery Date, Delivered to Client Date, Delivery Recorded Date	date as text	two date formats (`2-Jun-06` , `9/11/14`) plus text placeholders

Group	Columns	Type	Notes
Product	Product Group, Sub Classification, Item Description, Molecule/Test Type, Brand, Dosage, Dosage Form, Unit of Measure (Per Pack), First Line Designation	category	ARV 8,550 rows, HIV rapid tests (HRDT) 1,728; dosage missing in 1,736 rows
Vendor	Vendor, Manufacturing Site	category	73 vendors, 88 sites, mostly in India
Quantity and price	Line Item Quantity, Line Item Value, Pack Price, Unit Price	numeric	17 rows with zero value
Logistics cost	Weight (Kilograms), Freight Cost (USD), Line Item Insurance (USD)	mixed text / numeric	weight and freight mix numbers with notes; insurance missing in 287 rows

Why this dataset suits data mining: it has a clear target (late vs on time), a continuous target (freight cost), high-cardinality categories (country, vendor, molecule) for clustering and association rules, and a real data-cleaning problem in the cost columns.

Data quality and preprocessing

The ETL script (`scripts/01_clean_transform.py`) has been run on the full file and all 10,324 rows survive. The main problem is that the weight and freight columns mix numbers with notes. 2,445 rows point to another line item ("See ASN-93 (ID#:1281)"), and we resolve that pointer to get the number.

Issue found	Column(s)	Rows	Treatment
Cross-reference instead of a value	Weight, Freight Cost	2,445 each	Copy the value from the referenced ID; status = "Shared with ID"
Reference points to a row with no number	Weight / Freight Cost	126 / 105	Value null; status = "Shared - referenced value missing"
"Weight Captured Separately"	Weight	1,507	Value null, text kept as `weight_status`

Issue found	Column(s)	Rows	Treatment
"Freight Included in Commodity Cost"	Freight Cost	1,442	Value null, text kept as `freight_status` (cost is inside commodity price)
"Invoiced Separately"	Freight Cost	239	Value null, status kept
"N/A - From RDC" / "Date Not Captured"	PO Sent to Vendor Date	5,732	Null; lead time is only computed for Direct Drop orders
"Pre-PQ Process"	PQ First Sent to Client Date	2,681	Null
Two date formats in one column	all 5 date columns	all	Parsed with `%d-%b-%y`, then `%m/%d/%y`
Missing shipment mode	Shipment Mode	360	Category "Unknown"
Missing dosage	Dosage	1,736	"N/A" (test kits have no dosage)
Negative lead time	derived	5	Set to null (data-entry error)
Missing insurance	Line Item Insurance	287	Left null; excluded from insurance KPIs
Long vendor names	Vendor	73	`vendor_short` (text before the bracket, title case)

Engineered features

Feature	Formula	Used for
`delay_days`	Delivered date minus Scheduled date	RQ1, KPI cards
`is_late`	delay_days > 0	Classification target
`delivery_status`	Early / On time / Late 1-30d / Late >30d	Dashboard colour bands
`lead_time_days`	Delivered date minus PO sent date	RQ2
`pq_to_po_days`	PO sent minus PQ sent	RQ2, process time
`recording_lag_days`	Recorded date minus Delivered date	Data governance
`freight_per_kg`	Freight / Weight	RQ3
`freight_pct_of_value`	Freight / Line value	RQ3, cost efficiency

Feature	Formula	Used for
`total_landed_cost_usd`	Value + Freight + Insurance	RQ6
`units_total`	Packs x Units per pack	Volume KPIs
`region`	Country to region map	Slicers, clustering

Outputs: `data/processed/shipments_clean.csv` (flat), `data/processed/shipments_mongo.json` (nested, 13.4 MB), `data/powerbi/` star schema (one fact table, three dimension tables) and `data/processed/data_quality_report.csv` (the log above).

MongoDB design

The database `scms_lifeline` stores each line item as one nested document. The fields fall into natural groups: destination, fulfilment, vendor, product, dates, cost, logistics and KPIs. A document model fits this better than 33 flat columns. Summary collections are built from the main collection with `$merge`, so the dashboard reads small, pre-aggregated data.

Collections

Collection	Documents	Built by	Purpose
`shipments`	10,324	`insert_many` from `shipments_mongo.json`	Main fact store, nested, with a `$jsonSchema` validator
`vendors`	73	`$group` + `$merge`	Vendor scorecard: value, late rate, sites, countries served
`countries`	43	`$group` + `$merge`	Country risk: late rate, median delay, freight per kg
`monthly_kpis`	about 110	`$dateTrunc` + `$merge`	Time series for trend visuals
`ml_features`	10,324	`$project` + `$out`	Flat feature table for the Python mining step

Sample shipment document

```
{
  "_id": 3,
  "project_code": "108-VN-T01",
  "refs": { "pq": "Pre-PQ Process", "po_so": "SCMS-13", "asn_dn": "ASN-85" },
  "destination": { "country": "Vietnam", "region": "Asia", "managed_by": "PMO
- US" },
  "fulfilment": { "via": "Direct Drop", "inco_term": "EXW", "mode": "Air" },
  "vendor": { "name": "Aurobindo Pharma Limited", "manufacturing_site":
"Aurobindo Unit III, India" },
  "product": { "group": "ARV", "sub_class": "Pediatric", "molecule":
"Nevirapine",
               "dosage": "10mg/ml", "form": "Oral suspension",
"units_per_pack": 240, "first_line": true },
  "dates": { "scheduled": ISODate("2006-11-14"), "delivered": ISODate("2006-
11-14") },
  "quantity": { "packs": 1000, "units": 240000 },
  "cost": { "line_value": 6200, "unit_price": 0.03, "freight": 4521.5,
"freight_status": "Recorded" },
  "logistics": { "weight_kg": 358, "freight_per_kg": 12.63 },
  "kpi": { "delay_days": 0, "is_late": false, "status": "On time" }
}
```

Indexes (in `scripts/02_load_mongodb.py`)

- `{destination.country: 1, dates.delivered: -1}` for country and date filters
- `{fulfilment.mode: 1, kpi.is_late: 1}` for the mode reliability pipeline
- `{vendor.name: 1}`, `{product.group: 1, product.molecule: 1}`, `{dates.delivered: 1}`
- A text index on `product.description` and `vendor.name` for search

Aggregation pipelines (`mongodb/aggregations.js`)

#	Question	Main stages
Q1	Headline KPIs (value, freight, on-time %)	`$group`, `$addToSet`, `$project`
Q2	Late rate by mode x route	`$group` on compound key, `$match` n >= 30
Q3	Worst countries, 90th-percentile delay	`$percentile`
Q4	Lead time by vendor (Direct Drop)	`$match`, `$median`
Q5	Weighted freight per kg by mode x region	`$group`, `$divide`
Q6	How freight is recorded	`$sortByCount`

#	Question	Main stages
Q7	Vendor Pareto (cumulative share)	`$setWindowFields`, `$rank`
Q8	Spend by product group x year	`$year`, `$group`
Q9	Late-rate trend with 3-year moving average	`$setWindowFields` window [-2, 0]
Q10	Unit-price spread for the same molecule across vendors	two-level `$group`, `$push`
Q11	Flat feature table for mining	`$project`, `$out`

Power BI connection: use the MongoDB Atlas SQL / ODBC connector, or export the summary collections with `mongoexport --type=csv`. The repo also ships ready CSVs in `data/powerbi/`, so the dashboard can be built before MongoDB is set up.

Requirements and status: `$median`, `$percentile` and `$setWindowFields` need MongoDB 7.0 or later. The scripts are written but **not yet run against a live server**, so testing them is the first task in the MongoDB milestone.

Data mining methods and baseline results

All four methods in `scripts/03_data_mining.py` have been run on the cleaned data, and the numbers below are first-pass baselines. Freight cost can be predicted well (R^2 0.79). Lateness is harder: ROC-AUC is 0.72 on future years, just under the 0.75 target, so the classifier needs more work.

A. Classification: will a line item arrive late? (RQ1)

- **Target:** `is_late` (11.5% positive overall).
- **Features known when the order is placed:** country, mode, fulfilment route, INCO term, vendor, product group, sub-class, dosage form, scheduled month and year, log quantity, log value, unit price, first-line flag.
- **Validation:** time-based split. We train on schedules before 2013 (6,507 rows, 9.3% late) and test on 2013 to 2015 (3,817 rows, 15.3% late), so the model never sees the future.

Model	ROC-AUC (time split)	Precision (late)	Recall (late)
Logistic regression (balanced)	0.61	0.20	0.95
Gradient boosting (HistGB, balanced)	**0.72**	0.33	0.20

A random 75/25 split gives the gradient-boosting model an AUC of 0.887. That number is too optimistic because it mixes years. The drop to 0.72 shows the late rate shifts over time (9.3% to 15.3%). Next steps are a probability threshold set by cost, lag features (such as each vendor's late rate in the previous 6 months) and SHAP explanations.

B. Regression: what will freight cost? (RQ3)

- **Data:** 8,181 line items with positive freight and weight.
- **Target:** log(freight USD). **Features:** log weight, mode, country, route, INCO term, product group, vendor, log value, year.

Model	MAE (USD)	R^2
Baseline: median freight per kg of the mode x weight	11,569	-0.91
Gradient boosting	**3,840**	**0.79**

The model cuts error by two-thirds against the baseline (median freight on the test set is US$ 6,810). The baseline does worse than predicting the mean, which shows that freight is not linear in weight: big consignments get much cheaper rates per kg.

C. Clustering: country segments (RQ4)

We ran K-means on 24 countries with at least 20 line items. The features were late rate, 75th-percentile delay, air share, RDC share and log median freight-as-%-of-value, all standardised. The best silhouette score was at k = 2 (0.60), but that split only separates Burundi, so we used k = 4 (silhouette 0.37).

Segment	Countries	Late rate	Air share	Median freight / value
Reliable air-served	14 (e.g. Nigeria, Haiti, Vietnam, Kenya, Rwanda)	7.3%	90%	24%

Segment	Countries	Late rate	Air share	Median freight / value
High-volume truck hubs	7 (South Africa, Zambia, Mozambique, Tanzania, Uganda, Zimbabwe, Côte d'Ivoire)	13.5%	45%	9%
Fragile, high-cost	2 (Congo DRC, South Sudan)	18.9%	99%	188%
Outlier	Burundi	38.8%	100%	21%

D. Association rules: what goes with lateness? (RQ5)

We ran Apriori on six item types (mode, route, region, product group, sub-class, value tercile) plus a LATE item, with minimum support 1% and lift of at least 1.2. This gave 66 rules that point to LATE.

If a line item is…	Support	Confidence (late)	Lift
From RDC + Truck + high value	2.4%	26.2%	2.28
ARV + Truck + high value	2.5%	26.2%	2.28
Truck + high value	2.5%	25.9%	2.25
From RDC + Southern Africa + Truck	2.6%	24.3%	2.12
From RDC + ARV + Truck	4.3%	23.7%	2.07

Large trucked consignments from the regional warehouses are late more than twice as often as the 11.5% baseline. That is a concrete target for the logistics team.

Output files: `outputs/mining_results.json`, `country_clusters.csv`, `cluster_profiles.csv`, `association_rules_late.csv`.

Power BI dashboard design

The dashboard has five pages on a star schema: one fact table of 10,324 line items and four dimensions. Every page answers one research question, and the same slicers (year, region, country, mode, product group) sit on all of them. The model, measures and theme are ready in `powerbi/`. The `.pbix` will be built page by page in the next sessions.

Data model

Table	Rows	Key	Relationship
`fact_shipment`	10,324	id	many side of all relationships
`dim_country`	43	country_key	1 : many to fact; holds region
`dim_vendor`	73	vendor_key	1 : many to fact
`dim_product`	184	product_key	1 : many to fact; group, molecule, form, first-line
`dim_date`	2006 to 2015 calendar	Date	active to `delivered_date`, inactive to `scheduled_date`

Pages

Page	Question	KPI cards	Main visuals
1. Executive overview	How big is the programme and how well does it deliver?	Commodity Value, Shipments, Countries Served, On-Time Rate %, Freight % of Value	Value by year (column), filled map of value, late-rate trend line with previous year, donut of product groups
2. Delivery performance (RQ1, RQ2)	Where and why are shipments late?	Late Rate %, Avg Delay When Late, Severe Late (>30d), Median Lead Time	Matrix of mode x route late rate with conditional colour, delay distribution histogram (Early / On time / Late 1-30 / Late >30), country bar sorted by late rate, decomposition tree of late items by route > mode > country > vendor
3. Freight and cost (RQ3)	What drives freight cost?	Freight Cost, Freight per kg, Freight Recorded %	Scatter of weight vs freight (log axes) coloured by mode, freight per kg by mode x region, waterfall of landed cost (value + freight + insurance), freight-status breakdown

Page	Question	KPI cards	Main visuals
4. Vendor and product (RQ6)	Who do we depend on and are they good?	Vendor Value Share %, Vendor Score	Pareto chart (value bars + Cumulative Vendor Share % line), vendor scorecard table, unit price spread per molecule, treemap of molecules by value
5. Country risk and segments (RQ4, RQ5)	Which countries need safety stock?	Countries in each segment	Map coloured by cluster (from `outputs/country_clusters.csv`), segment profile table, top association rules table, what-if slider for safety-stock days

Key DAX measures (`powerbi/measures.dax`, 26 measures)

```
Late Rate % = DIVIDE ( [Late Line Items], [Line Items] )

Freight per kg =
VAR _rows = FILTER ( fact_shipment, fact_shipment[freight_usd] > 0 &&
fact_shipment[weight_kg] > 0 )
RETURN DIVIDE ( SUMX ( _rows, fact_shipment[freight_usd] ), SUMX ( _rows,
fact_shipment[weight_kg] ) )

Late Rate % PY = CALCULATE ( [Late Rate %], SAMEPERIODLASTYEAR (
dim_date[Date] ) )

Vendor Score (0-100) =
VAR _rel = [On-Time Rate %]
VAR _cost = 1 - MIN ( 1, [Freight % of Value] )
RETURN ROUND ( 100 * ( 0.6 * _rel + 0.4 * _cost ), 0 )
```

Other measures cover volume (Line Items, Shipments, Units), cost (Landed Cost, Freight % of Value), Pareto (Vendor Rank, Cumulative Vendor Share %) and the colour rule `Late Rate Colour` (red at 20% or more, amber at 12% or more, green otherwise).

Design rules: the theme file is `powerbi/lifeline_theme.json` (navy accent, grey for everything else; red, amber and green used only for status). There are at most 5 KPI cards per page, a slicer panel on the left, and every visual is titled with its finding, not its topic.

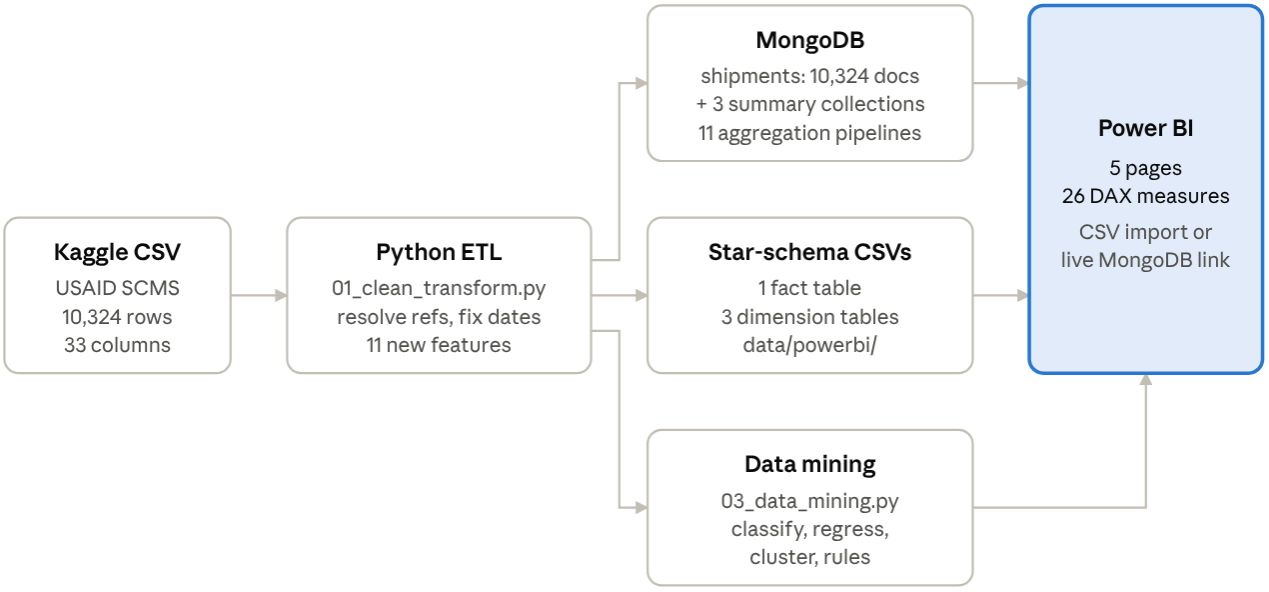
Loading: `powerbi/load_data.pq` has the Power Query scripts for the CSV star schema and a generated `dim_date`. Option B connects live to MongoDB through the Atlas SQL / ODBC connector.

System architecture and project structure

One ETL run creates every downstream input. Power BI can read the star-schema CSVs today and switch to a live MongoDB link once the database is set up.

One ETL run feeds MongoDB, Power BI and the mining models

Data flow left to right; script names as in the project folder



project data flow · 6 components

Project folder (`C:\Users\lenovo\Desktop\BDM-project`)

```
BDM-project/
├── README.md                       how to run everything, in order
├── requirements.txt                pandas, scikit-learn, pymongo, mlxtend
├── data/
|   ├── raw/SCMS_Delivery_History_Dataset.csv
|   ├── processed/                  shipments_clean.csv, shipments_mongo.json,
data_quality_report.csv
|   └── powerbi/                    fact_shipment.csv, dim_country.csv,
dim_vendor.csv, dim_product.csv
├── scripts/
|   ├── 01_clean_transform.py       ETL (tested)
|   ├── 02_load_mongodb.py          load + validator + indexes + summary
collections
|   └── 03_data_mining.py           classification, regression, clustering,
rules (tested)
├── mongodb/aggregations.js         11 pipelines (Q1-Q11)
├── powerbi/
|   ├── measures.dax                26 DAX measures
|   ├── load_data.pq                Power Query scripts
|   └── lifeline_theme.json         report theme
└── outputs/                        mining_results.json, clusters, association
rules
```

Hypotheses and early evidence

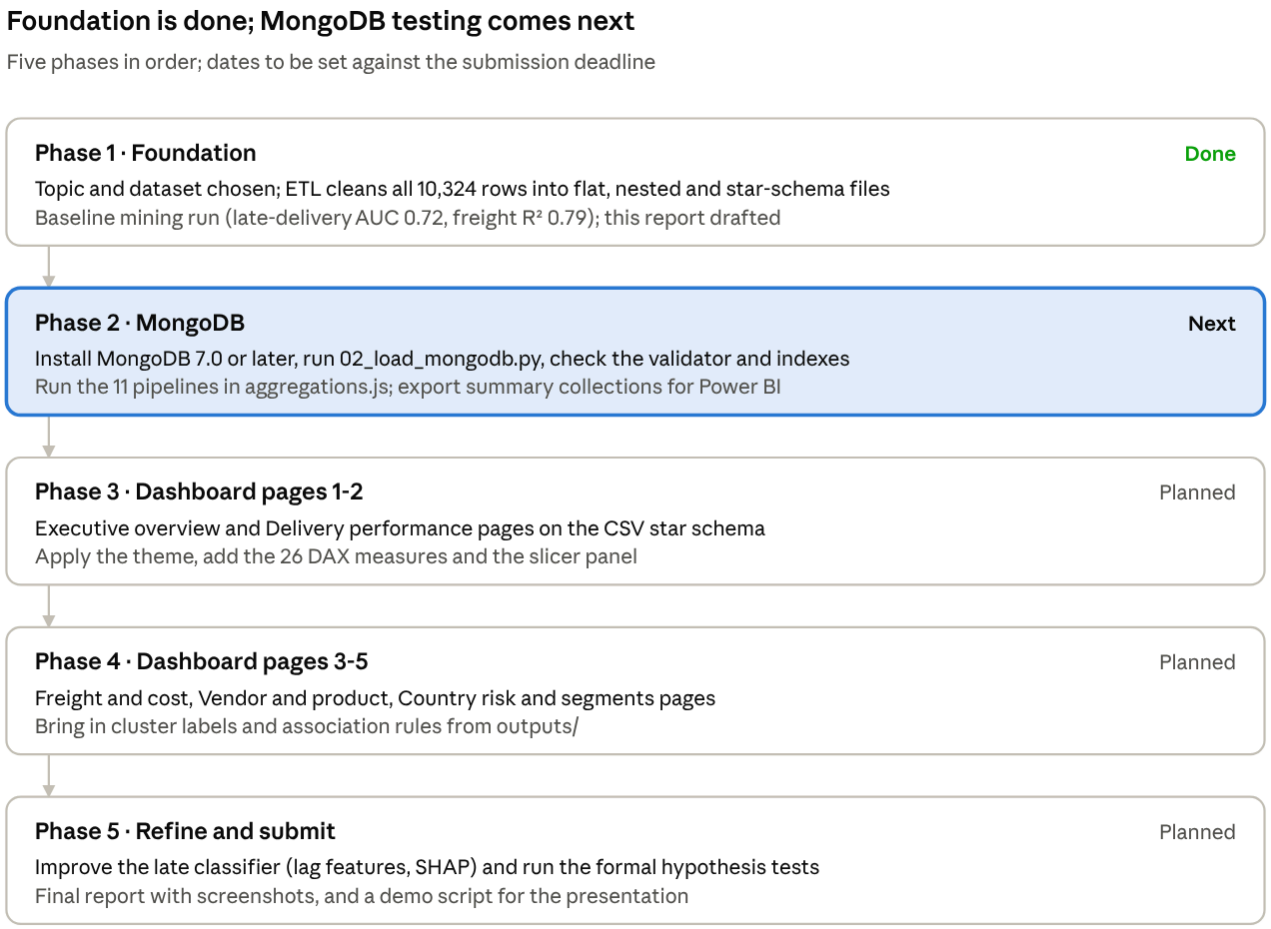
Five of the six hypotheses already have support from a first descriptive pass. The strongest finding is that lateness grew almost entirely on the regional-warehouse route: RDC orders went from 1.4% late in 2008 to 37.4% in 2014, while direct drops stayed between 2% and 8%. Formal tests (listed below) come in the analysis milestone.

#	Hypothesis	Early evidence from the data	Formal test	Dashboard page
H1	Direct Drop orders are more reliable than orders shipped from regional warehouses (RDC)	Late rate 5.3% vs 17.2%	Chi-square test; logistic regression coefficient	2
H2	The rise in lateness after 2009 comes from the RDC route	RDC late rate 1.4% (2008) to 37.4% (2014); Direct Drop 2-8% each year	Year x route interaction in logistic regression	1, 2

#	Hypothesis	Early evidence from the data	Formal test	Dashboard page
H3	Air is more reliable but costs much more per kg than truck or ocean	Late: Air 9.6%, Truck 16.1%, Ocean 17.5%. Median freight/kg: Air $9.35, Truck $1.82, Ocean $1.62	Kruskal-Wallis on freight/kg; chi-square on late	2, 3
H4	Freight per kg falls as consignment weight rises (economies of scale)	Log-log slope -0.53: doubling weight cuts cost per kg by about 31%	OLS on log(freight/kg) ~ log(weight) + mode	3
H5	Supplier spend is highly concentrated	RDC handles 66.7% of value; among the 72 direct vendors the top 5 hold 72.6%	Pareto curve, HHI index	4
H6	Fragile states cost far more to supply	Congo DRC and South Sudan: median freight is 188% of commodity value vs 9-24% elsewhere	Mann-Whitney U between segments	5

Roadmap and deliverables

Phase 1 is complete. The next working session will set up MongoDB, then the dashboard is built two pages at a time.



project roadmap · 5 phases

Deliverables checklist

- [x] ~~Unique topic and real Kaggle dataset chosen and downloaded~~
- [x] ~~ETL script and cleaned outputs (flat, nested JSON, star schema)~~
- [x] ~~Baseline data mining results~~
- [x] ~~Project report draft (this document)~~
- [ ] MongoDB loaded and all 11 pipelines verified
- [ ] Power BI pages 1-2 (Executive overview, Delivery performance)
- [ ] Power BI pages 3-5 (Freight and cost, Vendor and product, Country risk)
- [ ] Final report and presentation

Limitations, ethics and future scope

The biggest limitation is missing cost data: about 17% of freight values and 16% of weights are notes, not numbers, so cost findings only cover line items where they were recorded.

Limitations

- **Missing cost data:** freight is "Included in Commodity Cost" or "Invoiced Separately" in 1,681 rows, and weight is "Captured Separately" in 1,507. Cost KPIs are shown together with the `Freight Recorded %` measure so readers know how much data is behind them.
- **Shared consignments:** 2,445 line items take their weight and freight from another line in the same consignment. Summing them double-counts freight, so freight totals must deduplicate by ASN/DN in a later version.
- **No PO date for RDC orders:** lead time can only be measured for Direct Drop (4,920 rows).
- **Historic data:** 2006 to 2015 only. Routes, vendors and prices have changed since, so treat the findings as patterns, not current benchmarks.
- **Observational data:** association rules and model importances show correlation, not cause.
- **Unvalidated MongoDB layer:** the load and aggregation scripts have not yet been run on a live server.

Ethics

- The data has no personal or patient data. It is shipment-level and published openly by USAID.
- The licence is CC BY-ND 4.0. We cite the source and keep the raw file unchanged. Derived tables are for analysis only and are not republished as a new dataset.
- Country risk labels describe logistics performance, not the quality of a country's health system, and the report says so wherever segments are shown.

Future scope

- Add a forecast of monthly demand by molecule (Prophet or SARIMA) on `monthly_kpis`.
- Explain the classifier with SHAP and add a "risk score" column to the Power BI vendor table.
- Join external data (World Bank Logistics Performance Index, distance to port) to explain country effects.
- Publish the dashboard to Power BI Service with scheduled refresh from MongoDB Atlas.

References

1. Ardeshana, D. *Supply Chain Shipment Pricing Data*. Kaggle. https://www.kaggle.com/datasets/divyeshardeshana/supply-chain-shipment-pricing-data

2. USAID. *Supply Chain Shipment Pricing Data* (SCMS Delivery History). Catalogue entry: https://data.amerigeoss.org/dataset/supply-chain-shipment-pricing-data. Licence CC BY-ND 4.0.
3. Copy of the file used in this project (same 10,324 x 33 file, downloaded 5 Nov 2026): satoshi-F13/usaid-scms-dashboard
4. MongoDB Manual: Aggregation pipeline stages `$setWindowFields`, `$median`, `$percentile`, `$merge`. https://www.mongodb.com/docs/manual/reference/operator/aggregation-pipeline/
5. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
6. Raschka, S. (2018). MLxtend. *Journal of Open Source Software*, 3(24), 638.
7. Agrawal, R. and Srikant, R. (1994). Fast algorithms for mining association rules. *Proc. VLDB*, 487-499.
8. Microsoft. DAX function reference. https://learn.microsoft.com/dax/